



Report

Compilation of pan files

Bericht
Erzeugung von pan Files

Customer Auftraggeber	Goldi Solar Pvt.Ltd. Sankar, Ray Block No.149, Plot No. J & K1, Bs. IOC Petrol Pump,NH 8, Pipodara 394110 Dist. – Surat Indien
Test Report Prüfbericht	GSO002-003GSO0718-V01
Manufacturer / Type Hersteller / Typ	GOLDI SOLAR PVT. LTD. / GOLDI072F330PY24
Serial number Seriennummer	GG07180533702406
Internal ID / Order No. Interne ID / Auftragsnummer	GSO002 / 003GSO0718
Number of pages Seitenanzahl	4
Test date Datum der Prüfung	20.06.2018

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Date
Datum

01.07.2018

Head of testing laboratory
Leiter des Prüflaboratoriums

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Test object

Prüfgegenstand

Object Gegenstand	poly-Si PV module
Manufacturer Hersteller	GOLDI SOLAR PVT. LTD.
Type Typ	GOLDI072F330PY24
Cell material Zellmaterial	poly-Si
Serial number Seriennummer	GG07180533702406
Module area Modulfläche	1955 mm x 990 mm

Procedure

Verfahren

The pan file was created in PVsyst version 6.70. Input parameters for irradiance and temperature dependence were determined by indoor measurement of the IV characteristics of the PV module at a set of irradiance and temperature conditions (irradiance between 100-1100 W/m² in 100 W steps and temperature of 15, 25, 47, 50, 75 °C) according to IEC 61853-1. Measurements were carried out in a class AAA sun simulator. The detailed measurement results were reported in a separate report

For the "basic data" in PVsyst, the measured STC parameters were used instead of nominal values. The parameter T_k (pmpp) rel was extracted by a linear fit of the measured temperature dependence.

The model parameters R_p , R_s , R_p ($G_{inc}=0$) and exponential parameter were adjusted to optimize the agreement of modeled and measured efficiency for the complete range of temperatures and irradiances.

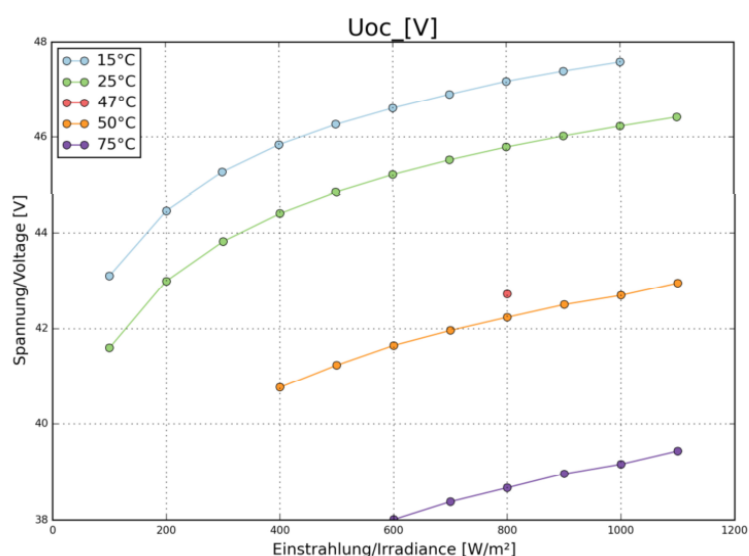
Results

Ergebnisse

According to IEC61853-1, for performance qualification testing three modules shall be selected at random from a production batch. The pan file "003GSO00718.pan" was created based on the measurement results for one module as reported in "GOG012_2_RAT_V01". Determined model parameters:

Model parameter	Value
Rp [ohm]	500
Rs [ohm]	0.32
Rp (Ginc=0)	2500
exp. param.	5.5
Tk (Pmpp) rel	0.38

Comparison of measured and modelled efficiency for the set of irradiance and temperature conditions specified in IEC61853-1:



Deviation of modelled and measured efficiency:

G [W/m²]	Deviation η [%]			
	15°C	25°C	50°C	75°C
1000	0.0%	0.0%	0.0%	0.0%
900	-0.1%	-0.1%	0.3%	0.3%
800	-0.1%	-0.1%	0.3%	0.3%
700	0.1%	-0.1%	0.2%	0.4%
600	-0.1%	-0.1%	0.4%	0.5%
500	-0.1%	-0.2%	0.3%	
400	-0.2%	-0.2%	0.2%	
300	-0.3%	-0.3%		
200	-0.6%	-0.5%		
100	0.3%	0.4%		

Results for reference power conditions according to IEC61853-1 and the corresponding IV curves were extracted from report "GOG012_2_RAT_V01".

Reference power conditions

Referenz-Bedingungen

	G [W/m ²]	T [°C]	Isc [A]	Uoc [V]	Impp [A]	Umpp [V]	Pmpp [W]	FF [%]	η [%]
STC <i>Standard test condition</i>	1000	25	9.176	46.231	8.700	38.148	331.888	78.232	17.148
LIC <i>Low irradiance condition</i>	200	25	1.826	43.007	1.710	36.650	62.690	79.848	16.195
HTC <i>High temperature condition</i>	1000	75	9.405	39.162	8.709	30.643	266.869	72.454	13.788
LTC <i>Low temperature condition</i>	500	15	4.547	46.267	4.319	39.366	170.025	80.817	17.570
NOCT <i>Nominal operating cell temperature</i>	800	47	7.412	42.729	6.971	34.864	243.041	76.740	15.697

IV curve

IU Kennlinie

